

Air Handling Unit (AHU): Process, Flow & Optimization Overview

Purpose:

AHUs condition and circulate air as part of an HVAC system. They regulate temperature, humidity, and air quality through a combination of cooling/heating coils, filters, fans, and dampers.

System Start-to-Finish Flow:

1. ****Air Intake & Filtration****
 - Outside air enters through louvers and dampers, mixed with return air.
 - Passes through pre-filters and secondary filters (MERV-rated).
2. ****Heating or Cooling Coil Interaction****
 - Air passes across hot water, steam, or chilled water coils.
 - Coil temps are modulated by BAS via control valves (0–100% open).
3. ****Humidity & Dehumidification****
 - Optional humidifiers inject steam or atomized water.
 - Cooling coils condense excess moisture (latent load removal).
4. ****Fan & Blower Section****
 - Belt or direct-drive fans push air to ducts. VFDs optimize energy use.
 - Pressure sensors adjust speed based on duct static pressure.
5. ****Supply Air Distribution****
 - Conditioned air is sent to zones via ductwork (velocity: 800–1,500 FPM).
 - Dampers and VAV boxes control delivery by zone.

Key Components & Definitions:

- ****Coil****: Transfers heat or cold to the air.
- ****Filter Rack****: Holds air filters to trap particulates.
- ****Blower Fan****: Drives air movement across all AHU sections.
- ****Mixing Box****: Combines return air and outside air.
- ****Access Panels****: For maintenance of internal sections.

Example Flow:

> A 10,000 CFM AHU pulls in 2,500 CFM of outdoor air, mixes with 7,500 CFM return. Filters remove particulates before air passes over 44°F chilled coil. VFD fan supplies air to four zones via VAVs, maintaining 72°F setpoint with $\pm 2^\circ\text{F}$ tolerance.